Protein Design Workshop 2026:

Design of mini-Protein Inhibitors Against SARS–CoV–2 Spike Protein by Artificial Intelligence-Based Tools

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Introduction

The COVID-19 pandemic was responsible for more than 7 million deaths worldwide and caused an unprecedented impact on the global economy, not seen since the Great Depression of 1929.[1,2] In this context, the rapid development of diagnostic tools and antiviral therapies to mitigate viral spread was essential to reduce mortality rates.[3] In this workshop, artificial intelligence (AI)-based protein design tools will be used to design mini-protein inhibitors against SARS-CoV-2.

The interaction between the SARS-CoV-2 spike protein and the human Angiotensin-Converting Enzyme 2 (ACE2) receptor is fundamental for viral infection, as it initiates the membrane fusion process that culminates in the internalisation of the viral genetic material into the host cell, followed by viral replication.[4,5] Mechanically, conformational changes in the SARS-CoV-2 spike protein expose the Receptor-Binding Domain (RBD), which binds to ACE2 through two large alpha-helices (helices H1 and H2) and one short beta-sheet (S1) (Figure 1).[5] The predominant interactions are concentrated on both helices; while H1 is responsible for providing most of the hydrophilic interactions, H2 contributes a high number of hydrophobic contacts. Blocking ACE2-RBD interactions is one of the most widely explored strategies to prevent viral entry.[6]

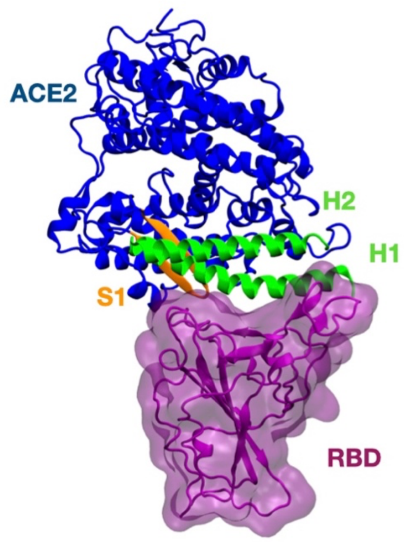

Figure 1: Structural representation of ACE2 (blue) bound to the RBD (purple) of the SARS-CoV-2 spike protein, with the two bundle helices (green) and the beta-sheet (orange) of the binding motif highlighted.

Rational protein design has played a major role in the development of mini-therapeutic proteins through structure-based knowledge and AI methods.[7] These strategies have inspired a new generation of antiviral therapies, including inhibitors, vaccines, and diagnostic tools. In 2020, such approaches were successfully applied to the *de novo design* of small and thermostable three-helix bundle (3HB) proteins capable of preventing SARS-CoV-2 infection by blocking interactions with the human ACE2 receptor with picomolar affinity.[8]

Inspired by the molecular details governing viral entry, many binders have been designed using similar approaches (Figure 2).

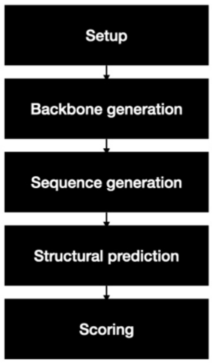

Figure 2: General *de novo* design pipeline. After defining the design approach, either for monomeric or multimeric proteins, a novel backbone topology is first created through “backbone generation”, followed by sequence design and structural refolding in “sequence generation” and “structural prediction”, respectively. Finally, the designs are ranked using an appropriate score function based on structural and thermodynamic properties.

In order to mimic the ACE2-RBD interactions, you should:

1. Select the ACE2 main binding motif (H1 and 2).
2. Maintain the binding residues unchanged during the design protocol.
3. Design a third alpha-helix backbone to stabilise the first two helices.
4. Design the side chains (sequence) based on the structure.
5. Refold the RBD-binder structure to validate the design.
6. Estimate the binding affinity with the aim of ranking the interfaces.

Methodology

Understanding the atomistic details of the ACE2-RBD interface is crucial for establishing successful design guidelines. Creating illustrative schemes and tables, as well as conducting an extensive literature review, is mandatory to define the design protocol. To support the designer throughout this process, several programs are essential to visualise and characterise proteins, such as PyMOL and the Rosetta program suite.[9,10] Characterising interfaces may require specific molecular modelling skills that will be briefly described here, although they are not the primary focus of this workshop, as these abilities are usually taught in other dedicated modules.

Selection of the ACE2 Binding Motif to Serve as a Basis for the Design Strategy

This stage of the design protocol may require previous experience with protein design in order to recognise the binding motif and understand how this choice can influence the subsequent steps. ACE2-RBD contacts occur between the RBD and H1, H2, and S1, with most interactions concentrated within the helices (Figure 3). Although these three structural elements together define a non-linear ACE2 binding motif, helices alone appear to provide sufficient structural information to design simpler helical proteins. Taking this into account, selecting fragment 30–82 is the best choice for the purpose of this workshop.

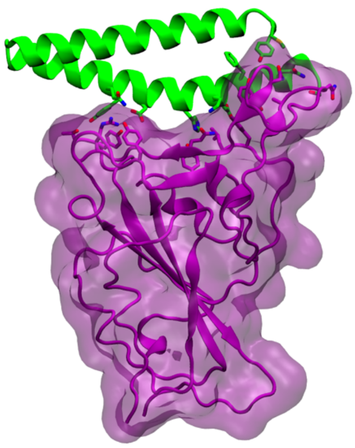

Figure 3: Illustrative scheme of the selected ACE2 binding motif (green) and the RBD (purple) of the SARS-CoV-2 spike protein. The interacting residues are displayed as tubes to highlight hydrophilic and hydrophobic interactions, such as hydrogen bonds, salt bridges, and hydrophobic contacts.

Why Is It a Good Starting Point to Preserve the Binding Residues During the Design Process?

Designing and ranking protein-protein interfaces are perhaps the most challenging aspects of binder design. Protein interfaces possess a peculiar chemical composition, displaying hydrophobic core-like features while simultaneously presenting hydrophilic interactions characteristic of protein-protein surfaces. Many approaches have failed to recognise and learn these ambiguous properties, making the development of machine learning (ML) algorithms one of the major focuses in the field. To circumvent these challenges, the ACE2 binding residues will initially be preserved in order to maintain the chemical environment of the interface.

Structure-Based Design of a 3HB Backbone Protein

Nowadays, most *de novo* protein design protocols follow a well-defined workflow consisting of the following steps:

1. **Backbone protein design**. The most popular software currently dedicated to designing protein frameworks is RoseTTAFold Diffusion (RFdiffusion).[11] Given an input structure and a set of instructions, the program diffuses backbone residues around the target and, through a denoising process, constructs protein backbone structures without side chains.

2. **Structure-based sequence design**. Sequence design algorithms generally follow two major approaches: force field-based and AI-based methods. To date, AI-based approaches have demonstrated higher experimental success rates and lower computational costs compared with force field-based ones. Among AI-based algorithms, ProteinMPNN is the most widely used for protein design.[12]

3. **Refolding**. Predicting structures from protein sequences is one of the most reliable ways to validate designs *in silico* before experimental validation. Among the available tools, AlphaFold3 (AF3) has revolutionised the field by predicting protein structures at atomistic resolution.[13] Although it may still fail for highly complex multimeric assemblies, AF3 remains one of the most powerful tools for validating designed binders by predicting their structures.

4. **Interface ranking**. Characterising and ranking interfaces may be one of the greatest challenges because this step defines the design quality and the ability of the binder to target the substrate. Assessing binding affinity and ranking designs accordingly helps prioritise candidates for experimental validation, thereby reducing costs. For this workshop, a protein-protein binding free-energy pipeline based on ML models trained on interface descriptors, called PBEE, will be used via Google Colab Notebook for this purpose.[14]

Hands-On Session

The complete protein design workflow (steps 1–4), including interface ranking, can be easily performed through the Diffusion (link here) and PBEE pipelines (link here), respectively.

Design ACE2-Like Proteins to Target the SARS-CoV-2 RBD

The design pipeline combines RFdiffusion, ProteinMPNN, and AF3 to generate backbone proteins, design sequences, and validate the resulting structures through a user-friendly and intuitive notebook (Figure 4). Through a series of control flags and a structural input, you will design four 3HB proteins.

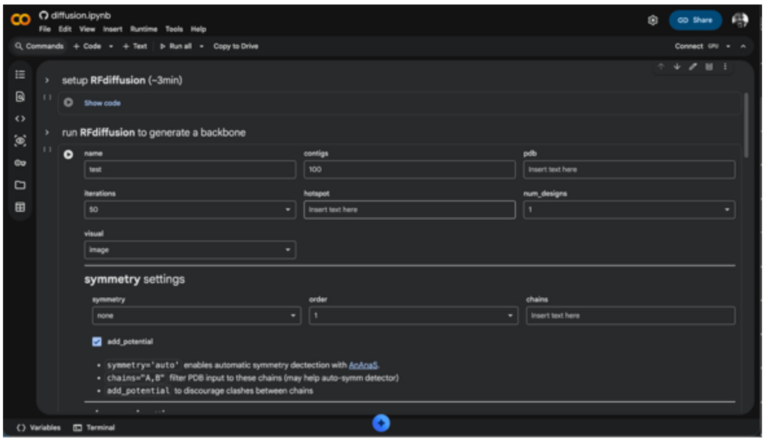

Figure 4: Print screen of the diffusion Google Collab Notebook to design proteins using *de novo* design approach.

To accomplish this, follow the steps below:

Define the structural input in the "pdb" string. As previously mentioned, structures with higher resolution generally result in designs with greater experimental success rates. Therefore, several criteria are important when selecting the best structural template. On the PDB website (link here), follow the procedure below to identify the most suitable structure for the design strategy:

1. In the search bar, use the following keywords to restrict the search to ACE2-RBD structures from SARS-CoV-2: ACE2 RBD SARS-CoV-2.

2. In the advanced search settings, select: "Experimental" under Structure Determination Methodology; "Homo sapiens" under Scientific Name of Source Organism; "Electron Microscopy" under Experimental Method; and "2.0–2.5 Å" under Refinement Resolution. A total of two structures should be listed.

3. Select the structure entitled "SARS-CoV-2 Omicron BF.7 RBD Complexed with Human ACE2", with PDB code 8WE1.

Define the base name in the "name" string. It is strongly recommended to use intuitive names that reflect the main objective, for example, "rbd-binder_model1". In this example, rbd-binder and model1 refer to the design of RBD binders and the use of the binding motif as a template, respectively.

Define the template structure and designed features in the "contigs" string. Since helices H1 and H2 will be used as the ACE2 binding motif to design 3HB binders against the RBD, you should select the RBD (chain B), the binding motif fragment (residues 19–83 from chain A), define the designed length (30–35 residues), and specify fragment connectivity (at the C-terminus). Therefore, the structure can be represented as "B:A19-83/30-35", where:

- B represents the RBD;
- :A19-83 defines residues 19–83 from chain A;
- While A19-83/30-35 defines the length of the designed extension at the C-terminus between 30 and 35 amino acids, :30-35/A19-83 defines elongation at the N-terminal.

In the next step, you should define the untouched residues in the "hotspot" string. To preserve the ACE2 (chain A) binding residues, these residues shall be defined in the hotspot string using syntax similar to the contig notation: "A24, A28, A31, A34, A38, A42, A79, A82, A83", as they are involved in hydrogen bonds and hydrophobic contacts.

Define the number of backbone designs. The required number of designs is highly dependent on the number of residues being designed. Although no strict rule exists, more than 20,000 designs are commonly required to generate proteins containing up to 80 amino acids, of which only dozens may ultimately proceed to experimental validation. Nevertheless, a large number of designs demands extensive computational resources, which are typically available only through High-Performance Computing (HPC) centres. For educational purposes and considering the limited workshop time, you should generate only 4 binders with 25 interactions each by selecting the respective numbers in the "num_designs" and "interactions" fields.

Set up the ProteinMPNN run. For relatively simple designs, the default configuration is usually sufficient to achieve satisfactory results. However, enabling the "initial_guess" option and increasing the number of recycles to 3 in the "num_recycles" field are strongly recommended for binder design.

Refold using AlphaFold. Since this workshop focuses on binder design, predicting the structure of the protein complex can provide not only information about the binder's propensity to adopt its functional conformation, but also clues regarding its ability to interact with the target protein. Such folding propensity is directly related to protein-protein interface structural predictions and not to protein-protein binding affinities. In order to activate multimer structural prediction, mark the "use_multimer" box.

Start the prediction by pressing "Run all" at the top of the webpage. Afterwards, a warning window will be displayed asking for authorisation. Press "Run anyway" to authorise it.

You should be able to track the whole process through images of backbone generation (Figure 5) and multimeric structure prediction, resulting from the sequence design process and AlphaFold prediction, in the “Display best result” section (Figure 6).

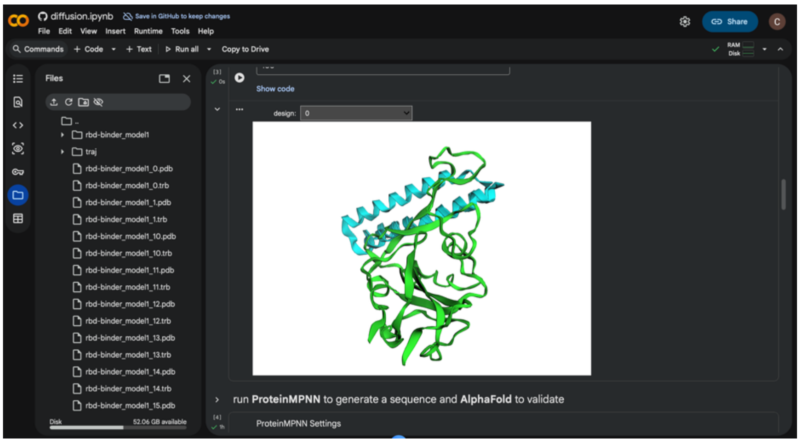

Figure 5: Backbone structure generated during the RFdiffusion process. RBD and designed binder are showed in green and blue, respectively.

Usually, good candidates should present a low root-mean-square deviation (RMSD) between the designed and predicted structures (RMSD < 2.0 Å), high prediction accuracy (pLDDT > 80), and high interface confidence (ipTM > 0.7). For binders, the interface is ranked based on the binding free energy (ΔG). Promising candidates may have a ΔG equivalent to or lower than that of the native complex (ΔG = -10.186 kcal/mol).

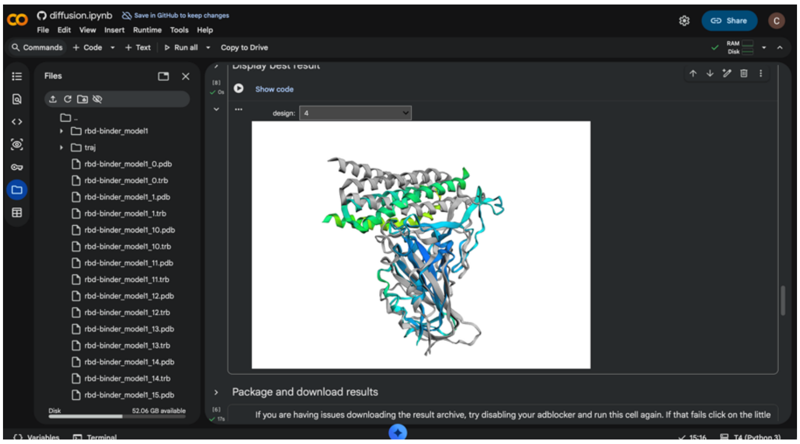

Figure 6: Comparison between designed and predicted structures. While the designed structures are displayed in grey, the predicted structures are shown according to the AF3 confidence colour map.

In the “Display best result” section, visualise your binders and compare the designed structures (grey) with the predicted ones (red–blue colour scale defining structural prediction accuracy). You will be asked to download the results.

Protein-Protein Binding Affinity Prediction

You will compare the binding energy of the best binder candidate identified during the structural prediction phase with the ACE2.

1. Access the PBEE notebook;
2. Run “Install all dependencies” to install PBEE on your google drive;
3. Upload the file best.pdb using “Choose Files”;
4. Change the “partner_1” and “partner_2” labels to A and B, respectively;
5. Mark the boxes “frcmod_struct” and “frcmod_scores”;
6. Run “Choose the setting and run”;
7. And compare the results with ACE2 binding affinity.

Explore your results

Describe your main results by answering the following questions:

1. Were the 3HB proteins successfully designed?
2. Do the binders interact with the SARS-CoV-2 RBD?
3. How similar are the designed structures to the predicted models?
4. Is the binding affinity comparable to that of ACE2?
5. Based on your observations, are four designs sufficient to obtain inhibitors with a high experimental success rate?
6. What additional design strategies could be implemented to improve the results?

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